EAMT-2026 · HANDS-ON TUTORIAL

Translation Evaluation *Tools for Everyone*

A Hands-On Tutorial for Freelance Translators and Smaller LSPs

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WELCOME

About this tutorial

Concise overview

of manual and automatic translation evaluation methods: what they are, what they measure, where each shines, and where each fails.

Hands-on with MATEO

a free, GDPR-compliant, no-code toolkit. Run BLEU, chrF, TER, and COMET on real multilingual data right from your browser.

Lightweight statistics

interpret your scores, compute confidence intervals and *p*-values, run correlation analysis, all using the tools you already have (Excel, plus a little help from LLMs).

3.5 hours total · 4 thematic parts · 30-minute coffee break after Part 2

About the presenter

A bridge between research and practice

- Professor at the University of Georgia (USA); affiliations span the Department of Philosophy, the Institute for Artificial Intelligence, and the Linguistics Program.
- Research interests: human and machine translation, MT evaluation, computational linguistics, philosophy of language and AI.
- ATA-certified English → Russian translator with 20 years of experience in the medical domain.
- Recently introduced the concept of Translation Analytics for freelancers (Balashov et al., 2025; 2026).
- Has taught graduate seminars on translation technologies, including individual student projects on translation analytics.
- Presented at ATA, AMTA, and MT Summit (2025), published in ACL, the ATA Chronicle, and on Slator.

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Human and Machine Translation: Cognitive, Linguistic, and Philosophical Perspectives

MOTIVATION

Why this tutorial?

THE GAP

Two separate worlds

MT research and big-tech industry workflows
rely heavily on automatic evaluation metrics — BLEU, chrF, TER, COMET, xCOMET, MetricX, etc. — to compare systems and track training progress.

Freelance translators and small LSPs
have traditionally relied on linguistic expertise and manual analysis. Automatic metrics were technically inaccessible, and seemingly irrelevant, to their daily work.

WHAT'S CHANGING

Two converging developments

No-code web toolkits
MATEO and other tools let non-programmers compute a suite of metrics through a browser. Zero installation, zero code.

LLMs as coding partners
Modern LLMs write simple, working analysis scripts on demand, even for users with only superficial programming background.

MOTIVATION

Why this tutorial?

WHAT'S CHANGING

New tasks requiring new skills

1. Freelancers are increasingly asked to do
“AI Content Strategy,” “Big Data Curation,” “QA Automation” — tasks that benefit directly from translation analytics.

2. New revenue streams
The skills we develop today are not just defensive (“keep up with AI”); they are **new revenue streams.**

WHAT WE'LL COVER TODAY

Tutorial schedule

1

45 min

Introduction to Translation Evaluation

Manual + automatic methods. Strengths, limitations, current developments.

2

45 min

Hands-On with MATEO

Walk-through of the interface. Run an evaluation on a real dataset.

COFFEE BREAK · 30 min

3

30 min

Interpreting Evaluation Results

Reading score tables, confidence intervals, sentence-level COMET in Excel.

4

45 min

Lightweight Statistical Analysis

Mean, variance, p-values, correlation. Excel + a touch of Python.

+ Final wrap-up Q&A (10–15 min)

WHAT WE'LL COVER TODAY

Tutorial schedule

Zouhar, V., Züfle, M., & Schmidtová, P. (2026). Human Evaluation of Multilingual Tasks. Tutorial at EAMT 2026.

1 — 45 min

Introduction to Translation Evaluation

Manual + automatic methods. Strengths, limitations, current developments.

2 — 45 min

Hands-On with MATEO

Walk-through of the interface. Run an evaluation on a real dataset.

COFFEE BREAK · 30 min

3 — 30 min

Interpreting Evaluation Results

Reading score tables, confidence intervals, sentence-level COMET in Excel.

4 — 45 min

Lightweight Statistical Analysis

Mean, variance, p-values, correlation. Excel + a touch of Python.

+ Final wrap-up Q&A (10–15 min)

1

Introduction to Translation Evaluation Methods

45 minutes · manual & automatic, with a tour of recent developments

Why evaluate translation quality?

> ***Is this translation any good?***
>
> *...compared to what, exactly?*

A reference translation

Most automatic metrics compare translation outputs against a human reference — your previous TM, an authoritative gold standard, or another professional output.

Another system's output

Pairwise comparisons let you rank engines, choose the best for a project, or quantify the gain from switching.

An expert's judgment

Human evaluators (you!) read and score outputs directly. Slow, expensive… But the gold standard.

Two big paradigms

MANUAL

Humans read & judge

Direct Assessment (DA)
rate sentences on a continuous or Likert scale

MQM (error annotation)
categorize and grade errors by severity

Ranking
rank multiple outputs against each other

AUTOMATIC

Algorithms compute scores

String-based
BLEU, chrF, TER — count word/character overlap with a reference

Neural-based
COMET, BLEURT, BERTScore — compare in semantic embedding space

LLM-as-judge
ask a large language model to score or annotate the translation

Manual evaluation: Direct Assessment

Score each translation on a numerical scale

- Continuous (0–100) or Likert-type (1–5, 1–7)
- Evaluators rate sentences for meaning preservation, fluency, or both.
- WMT Shared Tasks have used DA at large scale for over a decade.
- Recent variant: bucketed DA (HERMeS, VanHorn 2026): a coarse 4-bucket choice followed by a fine-grained ranking inside the bucket.

EXAMPLE SCALE

Academic 0.0 – 4.0

A	4.0	Excellent — publication-ready
B	3.0	Good — minor issues only
C	2.0	Acceptable — needs editing
D	1.0	Poor — substantial errors
F	0.0	Unusable / non-translation

Used in Balashov et al. (2025) for grading 1,080 MT/LLM segments

Manual evaluation: MQM

Don't just score — annotate the errors

Originally proposed by Lommel et al. (2014), MQM has become the de facto industry framework for fine-grained error annotation.

An evaluator marks each error span with:

- an error category (Accuracy, Fluency, Style, Terminology, …)
- a severity (Minor, Major, Critical)
- an optional comment justifying the call

Severities are weighted; the segment-level score is computed by summing penalties.

SEVERITY	PENALTY	EXAMPLE
Minor	1	Awkward register
Major	5	Wrong term in domain
Critical	25	Reverses meaning, safety risk

Resource cost

MQM campaigns require trained annotators and substantial time. Often impractical for small projects. But the conceptual framework (severity-weighted error annotation) is widely useful even when applied informally.

Manual evaluation: Ranking

Don't score — rank.

An evaluator sees several outputs side-by-side and orders them by quality.

- Easier than absolute scoring: you don't have to commit to a number.
- Captures relative quality reliably; absolute magnitudes are harder to recover.
- Pairwise comparisons aggregate well via TrueSkill, Bradley-Terry, etc.
- Used heavily in early WMT (Callison-Burch et al., 2007).

ONE SOURCE, FOUR OUTPUTS

1st	System B
2nd	System D
3rd	System A
4th	System C

Recent direct-assessment interfaces

Meet HERMeS =
Human Evaluation & Ranking of
Multiple Systems (VanHorn, 2026)

Two-step bucket + DA designs reduce cognitive load

STEP 1 · Choose a bucket

Best	*Sub-range: 22–28*
Good	*Sub-range: 15–21*
OK	*Sub-range: 8–14*
Poor	*Sub-range: 1–7*

→

STEP 2 · Refine within bucket

- 7-point integer scale within each bucket
- Final score on a 1–28 continuous range
- Coarse decision first → fine adjustment second
- Built-in randomization, blind presentation, integrity hashing

Automatic evaluation: why?

20
minutes
for COMET on a single 300-sentence document via MATEO

1080
segments
we manually graded for Balashov et al. (2025) — weeks of work

12 × 3
systems × language pairs
in our recent benchmark — humanly impossible to grade exhaustively

As data grows, detailed manual evaluation becomes unrealistic…

Automatic metrics let you scale evaluation to whole projects, run it overnight, and rerun it as systems change. They free human attention for the cases where it actually matters.

Two families of automatic metrics

STRING-BASED

Count overlap with the reference

BLEU · chrF · TER · METEOR · WER

Match words or characters between hypothesis and reference. Fast, cheap, deterministic

Pros

transparent, language-agnostic

Cons

blind to synonymy, paraphrase, and meaning

NEURAL-BASED

Compare in semantic space

COMET · BLEURT · BERTScore

Encode hypothesis and reference (and source) with a multilingual neural network; compare embeddings

Pros

captures meaning, robust to paraphrase, correlates better with humans

Cons

compute-heavy, opaque, model & training dependent

⇒ https://web.stanford.edu/~jurafsky/slp3/12.pdf

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BLEU — the original benchmark

Bilingual Evaluation Understudy · Papineni et al., 2002

- Modified *n*-gram precision: counts how many 1-, 2-, 3-, 4-grams in the candidate also appear in the reference, capped to avoid over-counting.
- Brevity penalty discounts candidates that are too short relative to the reference.
- Range 0–100. Higher is better. Document-level around 30 is typical for in-domain MT; 50+ is excellent.
- Sentence-level BLEU is noisy: design intent was always document-level.

⚠ Watch out

- *BLEU is sensitive to tokenization. For Japanese / Chinese, character-level tokenization is essential.*
- *Always report the sacreBLEU signature so others can reproduce your scores.*
- *Remember: never trust a single sentence-level BLEU.*

chrF — characters, not words

Character n-gram F-score · Popović, 2015

- Counts character (not word) *n*-grams shared between candidate and reference.
- Robust to morphologically rich languages (Russian, Finnish, Turkish) where small inflectional differences shouldn't break the metric.
- chrF2 weights recall twice as much as precision — improves correlation with human judgments.
- chrF++ adds word *n*-grams on top, blending both strengths.
- Range 0–100. Higher is better. In-domain MT typically falls in 50–70.

Practical tip

When BLEU and chrF disagree on the relative ranking of two systems, the morphology-sensitive chrF is usually the more trustworthy signal, especially for translations into Russian, German, or other inflection-heavy languages.

TER — edit distance for translators

Translation Edit Rate · Snover et al., 2006

Counts edits to transform the candidate into the reference.

Four edit operations:

- Insert · add a missing word
- Delete · remove an extra word
- Substitute · replace a wrong word
- Shift · move a phrase to a different position

TER = (number of edits) ÷ (reference length)

⚠

Lower is better!

TER measures effort. A score of 0 means no edits; a score of 100+ means rewriting from scratch. When you read your MATEO bar charts, remember: TER bars going DOWN is good news.

Closely related: HTER (human-targeted), HER (human edit rate)

COMET — the neural standard

Crosslingual Optimized Metric for Evaluation of Translation · Rei et al., 2020

HOW IT WORKS

SOURCE	HYPOTHESIS	REFERENCE
the original sentence	*the candidate translation*	*the human reference*

Multilingual neural encoder (XLM-R)

Single quality score · range ~0–100

Why it matters

- Trained on human ratings (WMT DA + MQM data)
- Strongest correlation with humans of any widely available metric
- **Slow… GPU strongly recommended**
- Scores not directly comparable across COMET versions (-22, -23)

Other neural metrics

BLEURT

Bilingual Evaluation Understudy with Representations from Transformers

Sellam, Das & Parikh (2020)

- Built on BERT, fine-tuned on synthetic and WMT human-judgment data.
- Reference-based, source not required.
- Comparable to COMET on many tasks but development has effectively stalled.

BERTScore

Token-level cosine similarity in BERT embedding space

Zhang et al. (2020)

- Each candidate token is matched to its most similar reference token.
- Aggregates into precision, recall, and F1 in semantic space.
- Generally outperformed by COMET on system-level rankings, but useful as a sanity check.

Current developments

xCOMET

Guerreiro et al., 2024

Goes beyond a single score:

- Span-level error detection
- MQM-style severity classification
- Source available

Useful when you need to know not just "is this bad?" but "what's wrong, where, and how seriously?"

MetricX-24

Juraska et al., 2024 · Google's WMT 2024 submission

Hybrid reference-based / reference-free:

- Two-stage training (DA, then MQM)
- Synthetic challenge data for failure modes
- State of the art on WMT 2024 metric tasks

Available, but not yet in MATEO. Run from the command line via Hugging Face.

LLM-as-judge

Just ask the LLM to score it.

PROMPT

```
You are an expert translation evaluator.  Source: "<English>".   Translation: "<German>".
Rate on 0-100 for adequacy.  Output only a number.
```

Strengths

- Strong system-level correlation with humans (Kocmi & Federmann, 2023)
- Can also produce error annotations (Lu et al., 2024)
- Just a prompt, no special training infrastructure

Cautions

- Sentence-level scores are noisier and harder to calibrate
- Cost: API calls add up over thousands of segments
- Privacy: cloud LLMs see your data, not OK for confidential work

Strengths and limitations at a glance

METRIC	TYPE	BEST FOR	WATCH OUT FOR
BLEU	string	fast doc-level rankings	tokenization, sentence-level noise
chrF	string	morphologically rich targets	still a surface metric
TER	string	post-editing effort estimate	lower is better!
COMET	neural	the default neural choice	compute cost, version drift
BLEURT	neural	cross-check on COMET	development stalled
BERTScore	neural	token-level similarity	weaker system rankings
xCOMET	neural	error spans + severity	newer, less established
MetricX-24	neural	WMT-grade rating accuracy	command-line only for now
LLM-as-judge	LLM	interpretable selective use	noise, cost, privacy

END OF PART 1

Quick Q&A - and a transition

Questions on Part 1?

Take 5 minutes to ask anything about manual or automatic evaluation methods.

BUT …

Translation evaluation is not only for "big tech!"

In Part 2 we'll meet MATEO — a free, no-code, browser-based toolkit that lets you run BLEU, chrF, TER, and COMET on your own data, with no installation, no programming, and no GPU.

2

Hands-On with MATEO

45 minutes · the no-code translation evaluation toolkit

Translation evaluation for everyone

Who actually needs this?

Choosing engines for a project

You have a 50K-word job and three NMT engines available. Which one wins on **this** domain? Run a small reference set and let the metrics decide.

Vetting translation memories

A client wants you to compare two candidate TMs from earlier projects. (i) Use them to pre-translate a sample in your CAT tool and evaluate the outputs with the metrics. (ii) Fine-tune/adapt your NMT engine with TM1 and TM2, then evaluate MT outputs with the metrics.

A new service offering

Translation analytics is increasingly billable. Adding rigorous, reproducible quality assessment to your portfolio sets you apart.

No-code toolkits

MATEO

BROWSER-BASED

Full BLEU/ chrF/ TER/ COMET/ BLEURT/ BERTScore. CLARIN- and HF-hosted. Today's main focus.

MutNMT

BROWSER-BASED

Erasmus+ multilingual NMT teaching tool, with its own evaluate function.

HuggingFace Spaces

HOSTED APPS

Search 'translation evaluation'. Quality varies; often spin-off MATEO clones.

Asiya / Asiya Online

BROWSER-BASED

Older toolkit with broader metric coverage. Less active today.

sacreBLEU

COMMAND-LINE

Reference implementation of BLEU/ chrF/ TER. No GUI; run via terminal or via a simple LLM-written script.

COMET CLI

COMMAND-LINE

Reference implementation. GPU strongly recommended. LLMs can write a wrapper for you.

MATEO

Free · no-code · comprehensive · GDPR-compliant

6

metrics in one toolkit

BLEU · chrF · TER · COMET · BLEURT · BERTScore

0

lines of code

browser-based; just upload your text files

≤1

MB upload limit

for the public CLARIN instance; clone it for more

Vanroy, Tezcan & Macken (2023)

Developed by the LT3 group at Ghent University, Belgium. Hosted as a Streamlit application on a GDPR-compliant server at the Instituut voor de Nederlandse Taal (CLARIN B Centre), with a mirror on Hugging Face. Open-source: clone, install locally, or deploy via Docker.

How it works under the hood

Your browser
upload aligned UTF-8 text

→

MATEO Streamlit app
orchestrates the metrics

→

CLARIN B server or HF
Computes the metrics

THE PRACTICAL UPSHOT

- All heavy computation (COMET, BLEURT, BERTScore) runs on remote resources.
- Your laptop just renders the page; even an old, modest one is fine.
- Speed depends on server load and internet, varies from minutes to half an hour.
- MATEO is open-source: clone, install locally, or run via Docker for more capacity.

Where to access MATEO

1. CLARIN B mirror (recommended)

`https://mateo.ivdnt.org`

GDPR-compliant, hosted by the Dutch Language Institute. Default for most users.

2. Hugging Face Space

`https://huggingface.co/spaces/BramVanroy/mateo-demo`

An identical mirror; useful when CLARIN is busy or unavailable. Clone it on your own HF Space!

3. Local install (advanced)

`github.com/BramVanroy/mateo-demo`

Clone and run with Streamlit (or Docker). For larger files (>1 MB) and full control.

Walking through MATEO: the input panel

mateo.ivdnt.org / Evaluate

Reference translation

reference_de.txt (~300 lines)

System outputs (up to 4)

system_1_de.txt

system_2_de.txt

system_3_de.txt

system_4_de.txt

Evaluate MT ▶

What you give it

- **ONE reference file**
  your gold-standard translation
- **UP TO FOUR system outputs**
  the candidate translations to compare
- **OPTIONALLY a source file**
  needed only if COMET-22 is selected
- **ALL files perfectly aligned**
  one sentence per line, same row order

Input requirements

Plain text, UTF-8

No .docx, no .pdf, no XLIFF, no inline tags. Save as .txt with UTF-8 encoding. Notepad++, Sublime, or your CAT tool's export-to-TXT all work.

One sentence per line

Lines correspond across files: line 47 of reference = line 47 of every output. Do not let your editor wrap long lines!

Identical line counts

If the reference has 302 lines, every system output must have exactly 302 lines. Check with Notepad++ (Windows) or Sublime (macOS).

Tip: open one of the .txt files from our repo in Notepad++ (or any text editor with UTF-8 support) to see the format!

Input requirements

Convert the uploaded TMX file to two perfectly aligned UTF-8 TXT files: src.txt and ref.txt. Requirements: exact same numbers of lines in both outputs. Each line in srx.txt must be perfectly aligned with the same-numbered line in ref.txt.

Plain text, UTF-8

No .docx, no .pdf, no XLIFF, no inline tags. Save as .txt with UTF-8 encoding. Notepad++, Sublime, or your CAT tool’s export-to-TXT all work.

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Selecting metrics

Tick what you want; leave the rest unchecked to save time.

- [x] **BLEU** — RECOMMENDED
  Fast (~30 sec per output)
- [x] **chrF** — RECOMMENDED
  Fast (~30 sec per output)
- [x] **TER** — RECOMMENDED
  Moderate (~2 min per output)
- [x] **COMET-22** — RECOMMENDED
  Slow (~15 min per output)
- [ ] **BLEURT**
  Slow (~10 min per output)
- [ ] **BERTScore**
  Moderate (~3 min per output)

My default for translators: BLEU + chrF + TER + COMET-22. That's already a rich picture!

Language-specific settings

BLEU and TER need the right tokenizer for Japanese & Chinese.

Default tokenization

...splits on whitespace.

Source:
私は本を読んでいます。

Tokens (default): 1

→ *The whole sentence becomes ONE "word." BLEU scores will be wildly inaccurate.*

Character tokenization

...treats each character as a token.

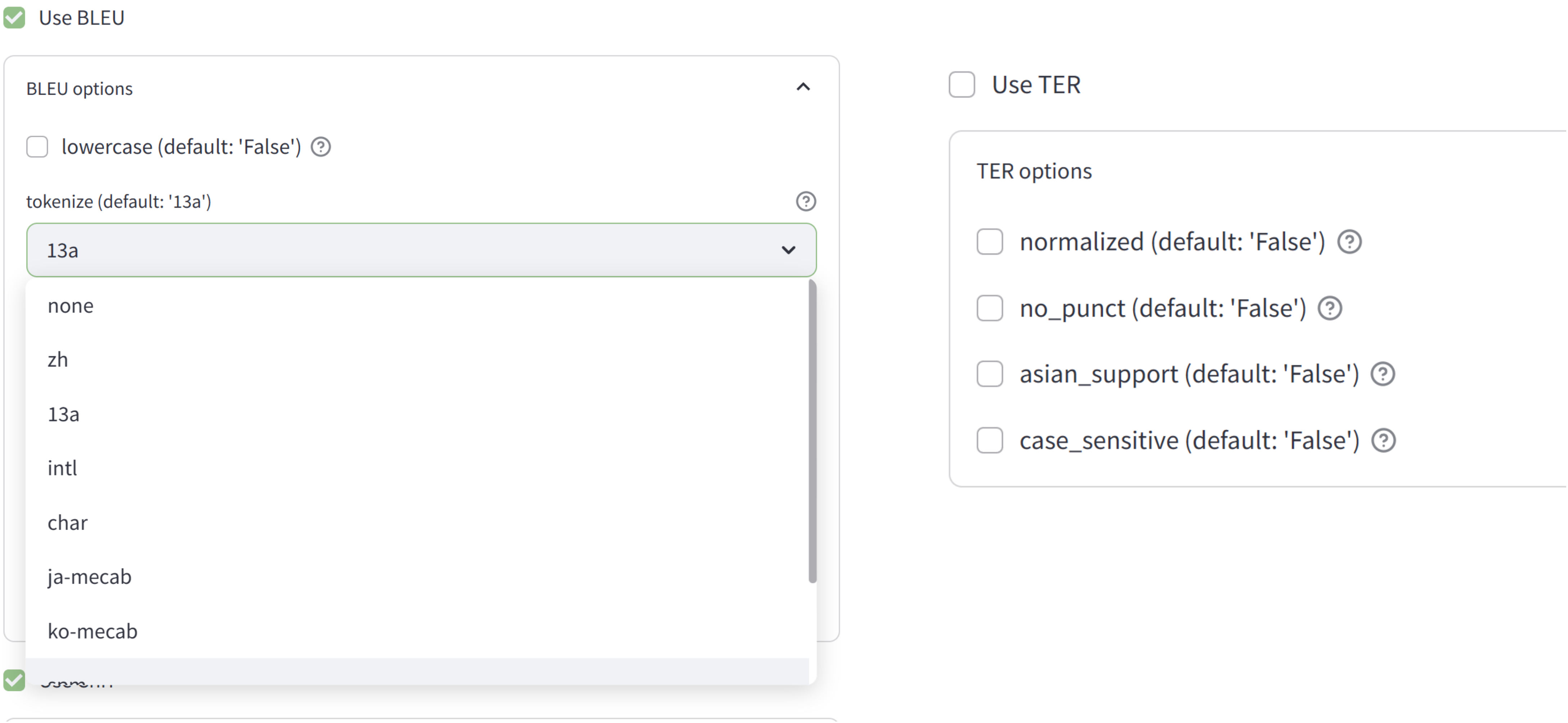
Source:
私 | は | 本 | を | 読 | ん | で | い | ま | す | 。

Tokens: 11

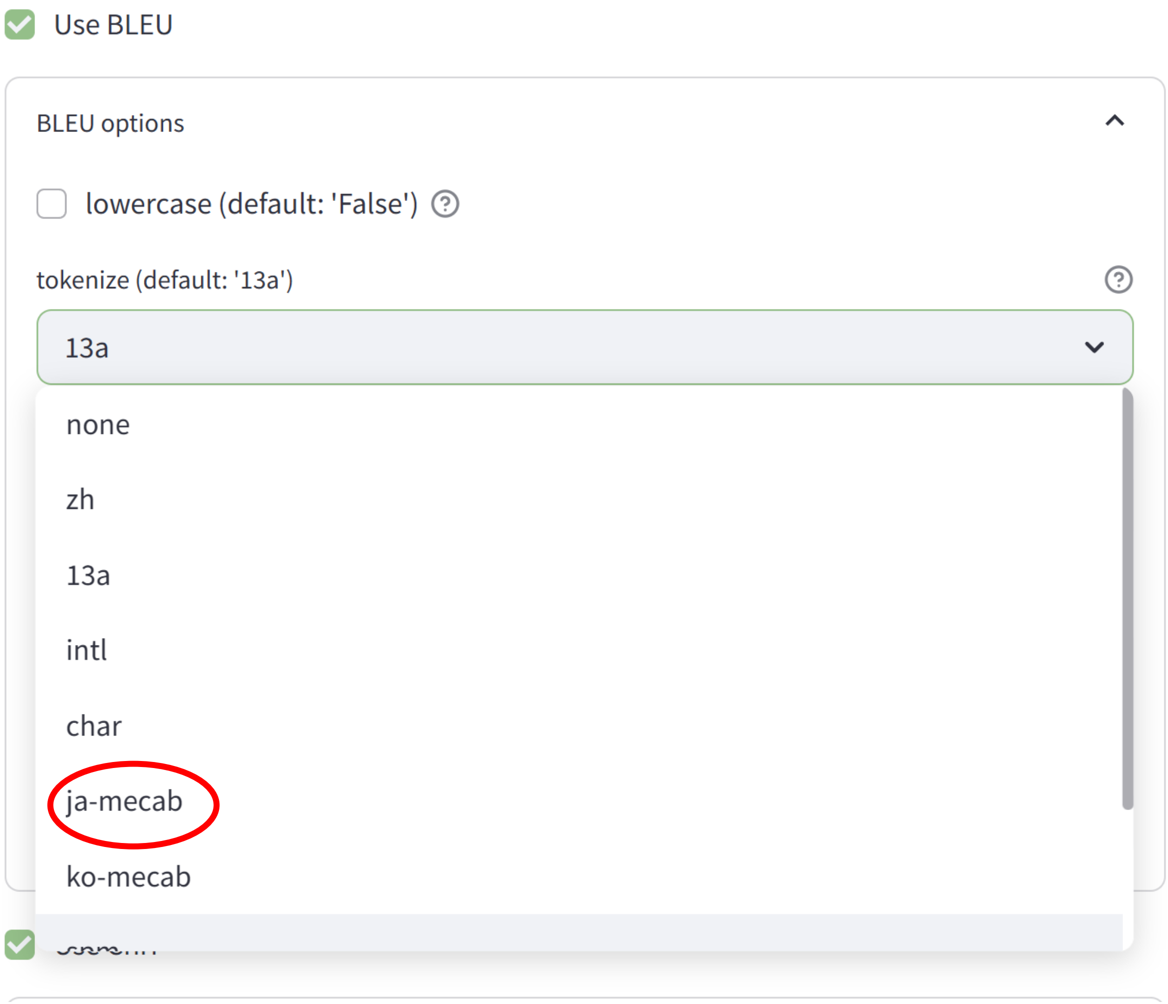
→ *MATEO's BLEU options for JA/ZH set this automatically when you pick the right target language settings.*

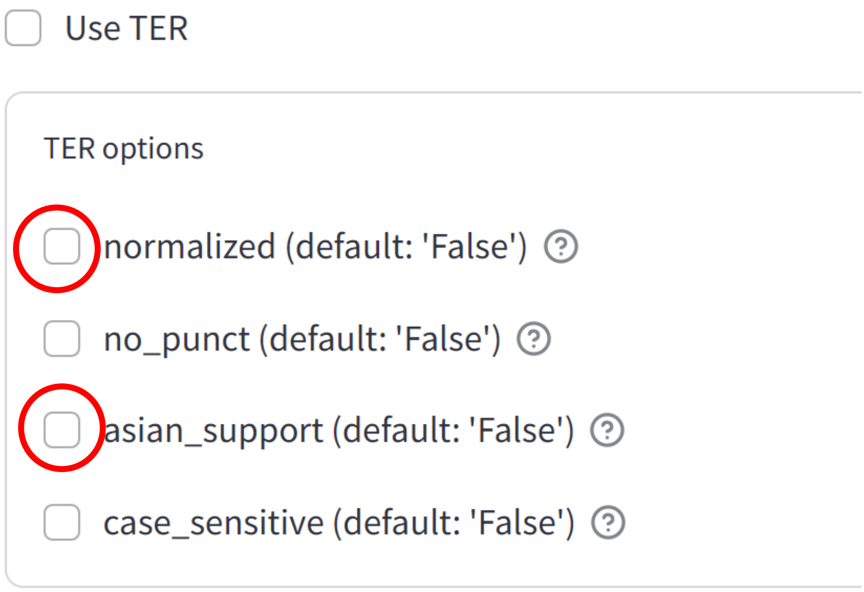
2.9a · TOKENIZATION FOR ASIAN LANGUAGES

Language-specific settings



Language-specific settings: ja





Language-specific settings: zh

- [x] Use BLEU

BLEU options

- [ ] lowercase (default: 'False')

tokenize (default: '13a')

13a

none
zh
13a
intl
char
ja-mecab
ko-mecab

- [ ] Use TER

TER options

- [ ] normalized (default: 'False')
- [ ] no_punct (default: 'False')
- [ ] asian_support (default: 'False')
- [ ] case_sensitive (default: 'False')

Language-specific settings

```
BLEU: nrefs:1|bs:1000|seed:12345|case:mixed|eff:no|tok:zh|smooth:exp|version:2.5.1|mateo:1.8.4
chrF2: nrefs:1|bs:1000|seed:12345|case:mixed|eff:yes|nc:6|nw:0|space:no|version:2.5.1|mateo:1.8.4
TER: nrefs:1|bs:1000|seed:12345|case:lc|tok:tercom|norm:yes|punct:yes|asian:yes|version:2.5.1|mateo:1.8.4
comet: nrefs:1|bs:1000|seed:12345|m:Unbabel/wmt22-comet-da|version:2.2.6|mateo:1.8.4
```

Language-specific settings

```
BLEU: nrefs:1|bs:1000|seed:12345|case:mixed|eff:no|tok:zh|smooth:exp|version:2.5.1|mateo:1.8.4
chrF2: nrefs:1|bs:1000|seed:12345|case:mixed|eff:yes|nc:6|nw:0|space:no|version:2.5.1|mateo:1.8.4
TER: nrefs:1|bs:1000|seed:12345|case:lc|tok:tercom|norm:yes|punct:yes|asian:yes|version:2.5.1|mateo:1.8.4
comet: nrefs:1|bs:1000|seed:12345|m:Unbabel/wmt22-comet-da|version:2.2.6|mateo:1.8.4
```

Language-specific settings

```
BLEU: nrefs:1|bs:1000|seed:12345|case:mixed|eff:no|tok:ja-mecab-0.996-
IPA|smooth:exp|version:2.5.1|mateo:1.8.4
chrF2: nrefs:1|bs:1000|seed:12345|case:mixed|eff:yes|nc:6|nw:0|space:no|version:2.5.1|mateo:1.8.4
TER: nrefs:1|bs:1000|seed:12345|case:lc|tok:tercom|norm:yes|punct:yes|asian:yes|version:2.5.1|mateo:1.8.4
comet: nrefs:1|bs:1000|seed:12345|m:Unbabel/wmt22-comet-da|version:2.2.6|mateo:1.8.4
```

Do we really need BLEURT and BERTScore?

Short answer: usually no.

COMET, BLEURT, and BERTScore all live in the same neural-embedding space and broadly agree.

- **Run COMET-22 by default.**

It is currently the best validated against human judgment and is the de facto industry standard.

- **Add BLEURT only if…**

you want a sanity check, or you suspect domain-specific issues with COMET.

- **Add BERTScore only if…**

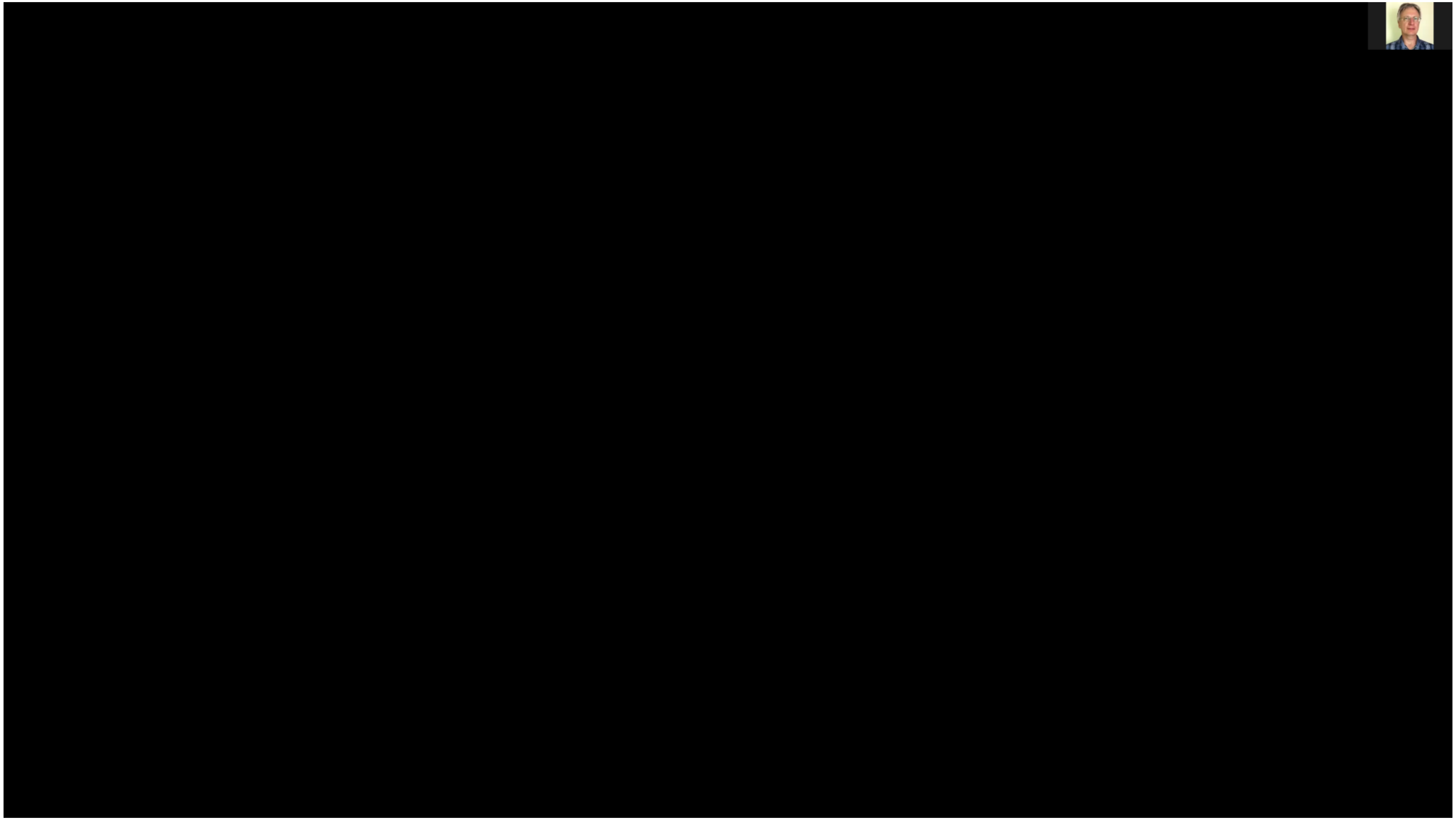
you specifically want token-level similarity rather than a single score per segment.

Time is finite

Each extra metric costs another 10+ minutes of wait time. Spend that time on more important things.

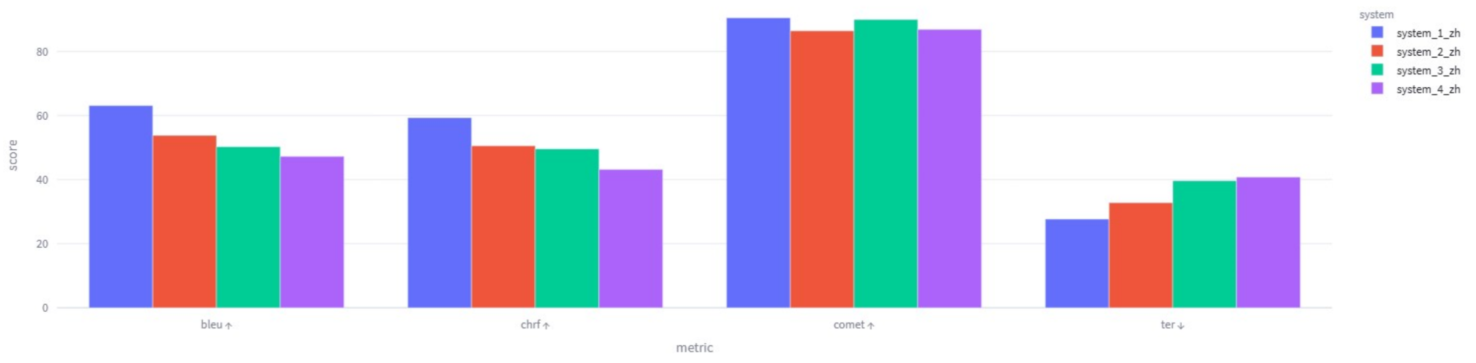
2.10.2

DEMO



Output: comparative charts

MATEO produces bar + radar charts per metric. Each bar = one of your candidate systems.



READING THE CHART

- **Higher = better**

(except TER — lower wins!)

- **Eyeball the gaps**

Tiny gaps may not be statistically significant — see Part 3.

- **Each metric tells its own story**

Don't expect rankings to be identical across BLEU / chrF / COMET. Disagreement is informative.

Output: score tables: `mateo-evaluation.xlsx`

MATEO downloads tables in Excel

system	comet	BLEU	chrF2	TER
Baseline: system_1_zh	90.51	63.11	59.31	27.68
system_2_zh	86.44	53.80	50.52	32.78
system_3_zh	90.00	50.29	49.59	39.61
system_4_zh	86.90	47.25	43.17	40.79

Output: score tables: `mateo-evaluation-ci.xlsx`

MATEO downloads tables in Excel — with optional 95% CIs and p-values

system	comet (μ ± 95% CI)	BLEU (μ ± 95% CI)	chrF2 (μ ± 95% CI)	TER (μ ± 95% CI)
Baseline: system_1_zh	90.5 ± 0.5	63.1 ± 2.3	59.3 ± 2.8	27.7 ± 2.0
system_2_zh	86.4 ± 0.9*	53.8 ± 2.4*	50.5 ± 2.8*	32.8 ± 2.0*
system_3_zh	90.0 ± 0.4*	50.3 ± 1.9*	49.6 ± 2.4*	39.7 ± 2.2*
system_4_zh	86.9 ± 0.7*	47.2 ± 2.1*	43.1 ± 2.5*	40.8 ± 2.1*

HOW TO READ THIS TABLE?

Output: score tables: `mateo-evaluation-ci.xlsx`

MATEO downloads tables in Excel — with optional 95% CIs and p-values.

SYSTEM	BLEU (μ ± CI)	chrF2 (μ ± CI)	TER (μ ± CI)	COMET (μ ± CI)
MT1	29.6 ± 6.0	52.8 ± 4.4	62.4 ± 5.3	83.7 ± 1.9
MT2	30.3 ± 5.7	54.6 ± 4.0	62.0 ± 4.8	**86.2 ± 1.7** *
MT3	**22.6 ± 4.5** *	**46.8 ± 3.8** *	**68.8 ± 4.4** *	**81.2 ± 2.0** *
MT4	**24.9 ± 4.7** *	**49.2 ± 3.8** *	**66.8 ± 4.5** *	82.9 ± 2.0

HOW TO READ THIS TABLE

μ ± CI = mean score ± 95% bootstrap confidence interval. When intervals overlap, the difference might be noise.

* = significantly different from the baseline (here MT1) at $p < 0.05$. No * means “could be chance.”

Output: sentence-level COMET: `mateo-sentences.xlsx`

MATEO also gives you a single Excel file with every sentence and its COMET score.

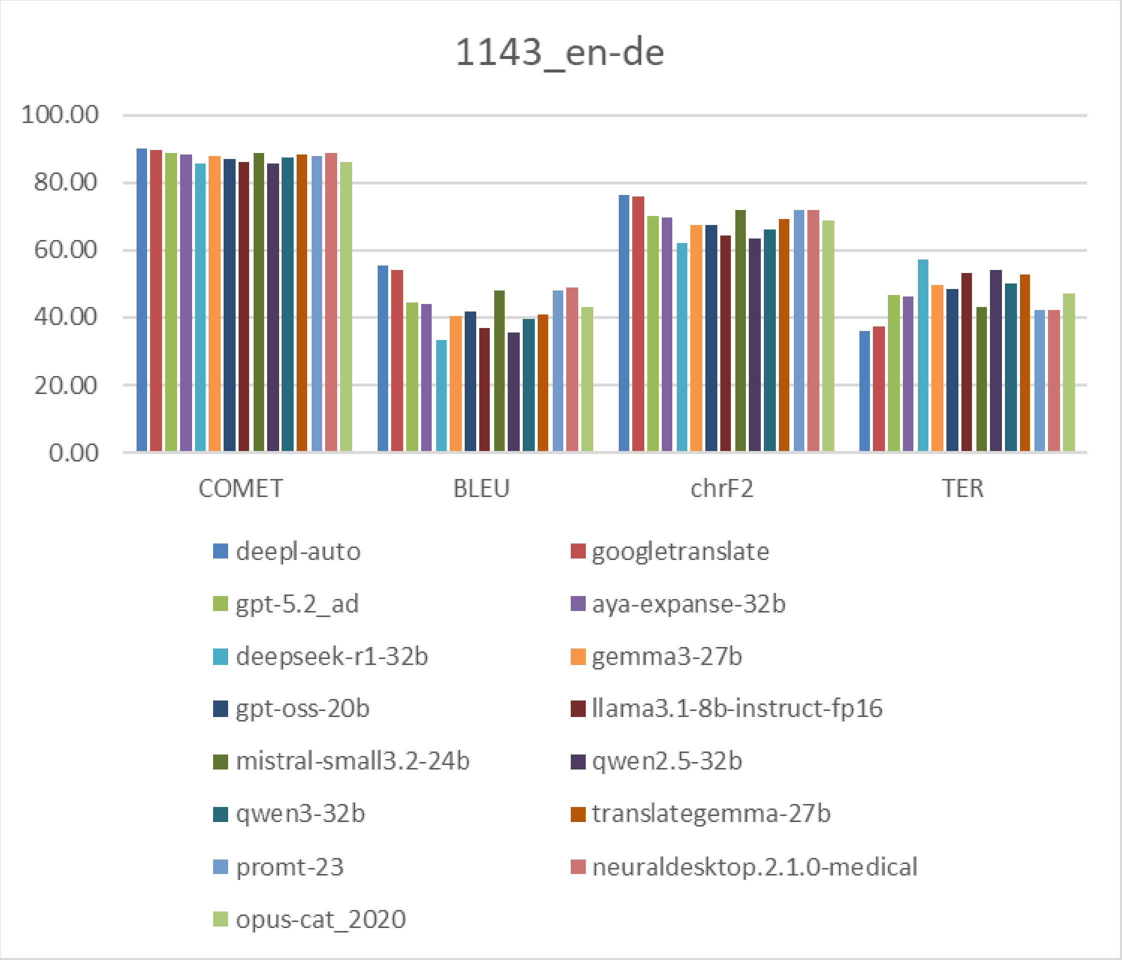
ID	SOURCE (EN)	REFERENCE (DE)	MT1 OUTPUT	MT1 COMET	MT2 OUTPUT	MT2 COMET
00042	Spinal cord injuries…	Rückenmarkverletzungen…	Verletzungen des Rückenmarks…	0.891	Rückenmark-Verletzungen…	0.876
00043	The patient was…	Der Patient wurde…	Der Patient war…	0.812	Der Patient wurde…	0.945
00044	Daily exercises help…	Tägliche Übungen…	Tägliche Übungen helfen…	0.923	Tägliche Übungen…	0.918
00045	Always consult your…	Konsultieren Sie immer…	Beraten Sie sich immer…	0.745	Konsultieren Sie stets…	0.889

Why this Excel file is gold

Document-level scores tell you which system is best on average. The Excel spreadsheet tells you which sentences each system handled well and which it botched. Filter, sort, run correlations, build your own charts. We'll dive deeper into all that in Parts 3 and 4.

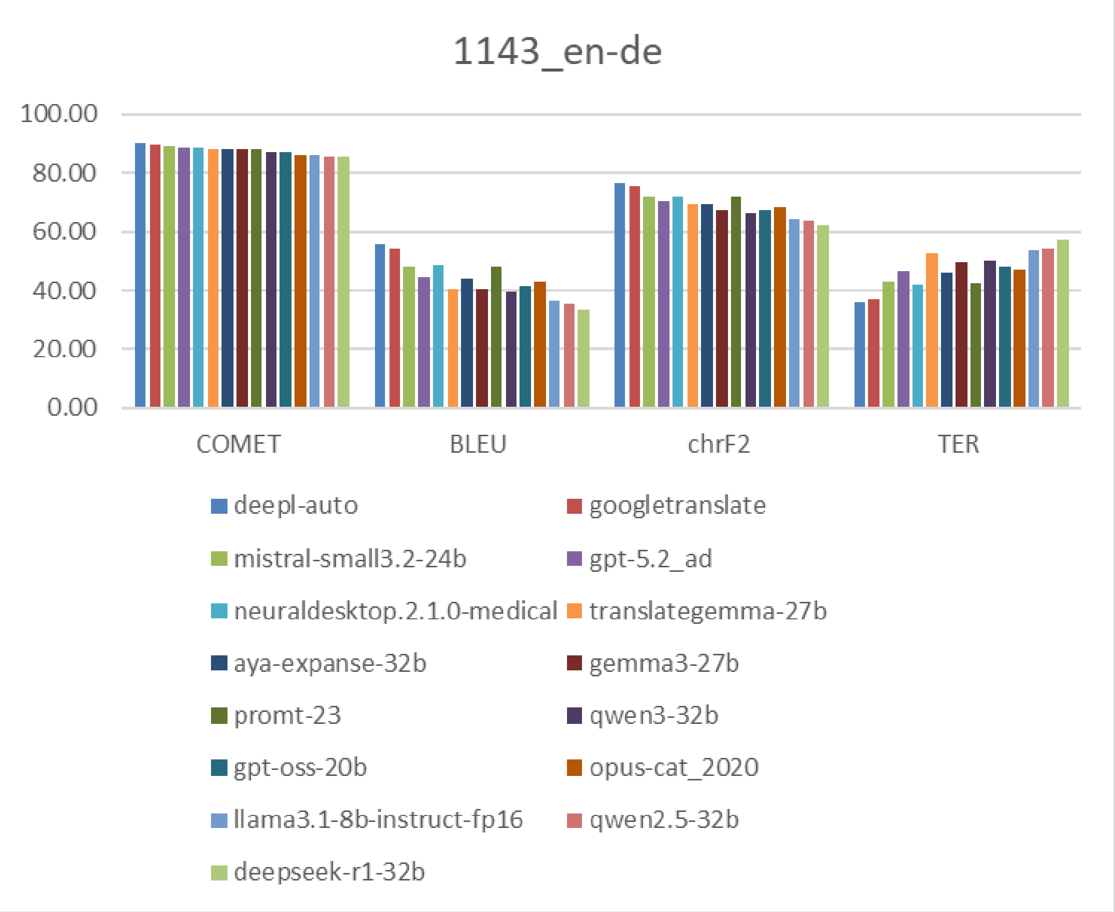
Output: compile your own tables and charts

	COMET	BLEU	chrF2	TER
deepl-auto	90.07	55.64	76.45	36.17
googletranslate	89.51	54.31	75.71	37.13
gpt-5.2_ad	88.75	44.42	70.26	46.79
aya-expanse-32b	88.18	44.13	69.46	46.15
deepseek-r1-32b	85.61	33.33	62.23	57.12
gemma3-27b	88.05	40.39	67.47	49.66
gpt-oss-20b	86.90	41.61	67.56	48.27
llama3.1-8b-instruct-fp16	86.23	36.65	64.36	53.43
mistral-small3.2-24b	88.91	47.98	71.88	43.10
qwen2.5-32b	85.75	35.48	63.57	54.07
qwen3-32b	87.23	39.43	66.26	50.31
translategemma-27b	88.31	40.67	69.25	52.69
promt-23	87.94	47.86	71.94	42.33
neuraldesktop.2.1.0-medical	88.68	48.80	71.73	42.20
opus-cat_2020	86.28	42.87	68.57	46.86



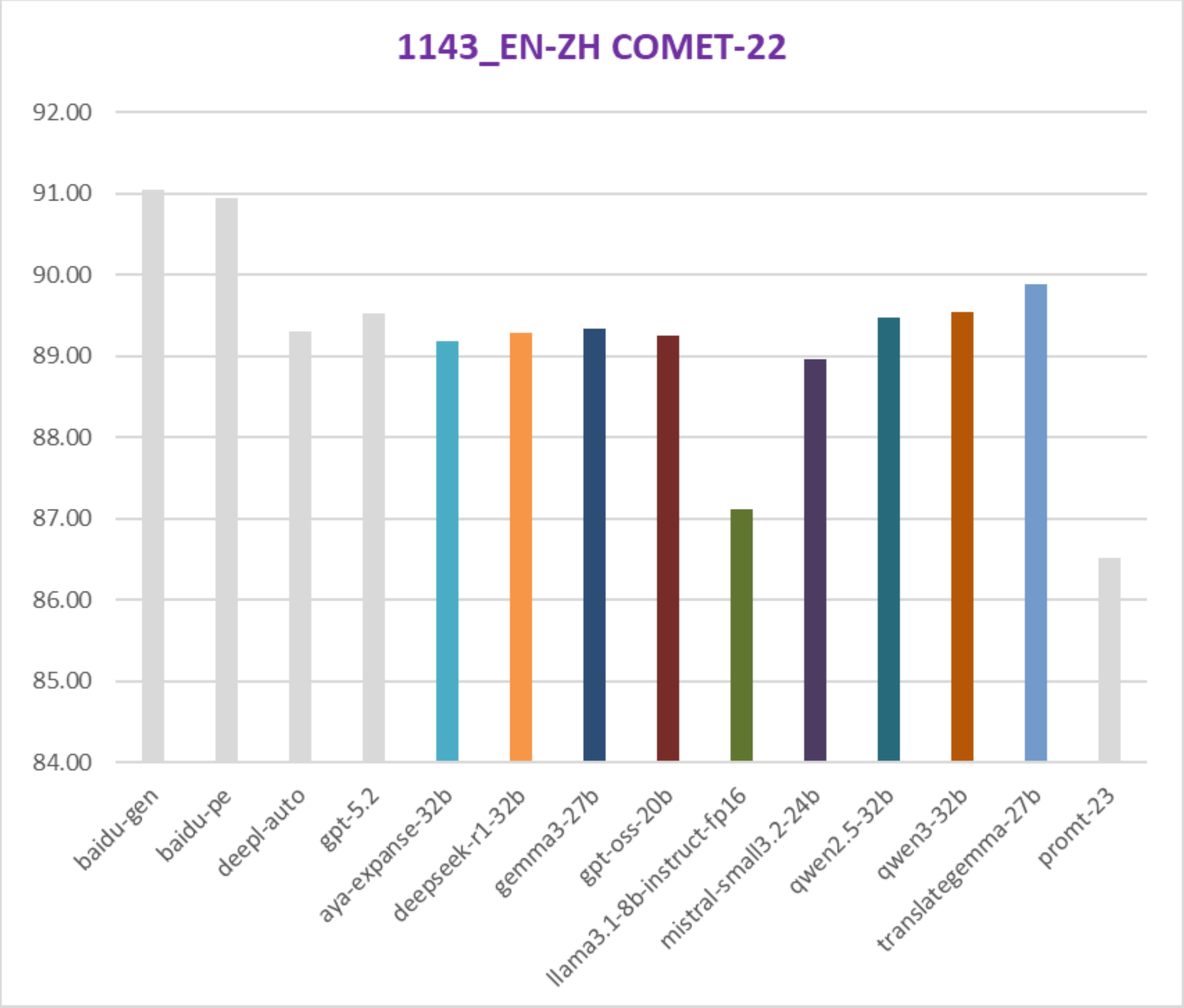
Output: compile your own tables and charts

Ranked by COMET scores				
	COMET	BLEU	chrF2	TER
deepl-auto	90.07	55.64	76.45	36.17
googletranslate	89.51	54.31	75.71	37.13
mistral-small3.2-24b	88.91	47.98	71.88	43.10
gpt-5.2_ad	88.75	44.42	70.26	46.79
neuraldesktop.2.1.0-medical	88.68	48.80	71.73	42.20
translategemma-27b	88.31	40.67	69.25	52.69
aya-expanse-32b	88.18	44.13	69.46	46.15
gemma3-27b	88.05	40.39	67.47	49.66
promt-23	87.94	47.86	71.94	42.33
qwen3-32b	87.23	39.43	66.26	50.31
gpt-oss-20b	86.90	41.61	67.56	48.27
opus-cat_2020	86.28	42.87	68.57	46.86
llama3.1-8b-instruct-fp16	86.23	36.65	64.36	53.43
qwen2.5-32b	85.75	35.48	63.57	54.07
deepseek-r1-32b	85.61	33.33	62.23	57.12



Output: compile your own tables and charts

	COMET
baidu-gen	91.04
baidu-pe	90.95
deepl-auto	89.31
gpt-5.2	89.53
aya-expanse-32b	89.19
deepseek-r1-32b	89.29
gemma3-27b	89.34
gpt-oss-20b	89.26
llama3.1-8b-instruct-fp16	87.11
mistral-small3.2-24b	88.96
qwen2.5-32b	89.48
qwen3-32b	89.55
translategemma-27b	89.88
promt-23	86.51



Output: compile your own tables and charts

Microsoft Excel
Worksheet

Hands-on: download an evaluation set

From the tutorial GitHub repository

`github.com/YuriBalashov/eamt2026-eval-tutorial`

DE	RU	JP	CN
EN → DE	**EN → RU**	**EN → JA**	**EN → ZH**
≈ 300 sentences	*≈ 300 sentences*	*≈ 300 sentences*	*≈ 300 sentences*
Download .zip	Download .zip	Download .zip	Download .zip

All sets are subsets of the Reeve Foundation Multilingual Corpus (RFMC), a real medical-domain translation project, with system outputs from commercial NMT engines and locally-runnable LLMs from our recent benchmarks (Balashov et al., 2025; 2026).

Quick Q&A on Part 2?

Take 5 minutes to ask anything about MATEO.

Run it — and let it work over the break

If you didn't have a chance to try MATEO before the tutorial, try it now! ⇒

1 Upload the files

Reference + 3-4 system outputs + (optionally) source. Drag & drop into MATEO.

2 Tick BLEU + chrF + TER + COMET-22

These four are our default. For Asian languages, set the BLEU tokenize option to 'ja-mecab' or 'zh', and TER to 'asian_support' and 'no_punct'.

3 Click "Evaluate MT" and step away (and hold your breath…)

It will work for ≈ 15-25 minutes (server load dependent). Perfect timing for coffee.

Or else: use the MATEO outputs from our GitHub repo (last slide)

BEFORE YOU LEAVE MATEO website: download the bar charts (PNG) and all the score/sentence tables (Excel) so we have them in Part 3.

Coffee break

30 minutes

3

Interpreting Evaluation Results

30 minutes · reading what MATEO gave you

Welcome back

By now you should have:

MATEO bar charts

PNG files showing BLEU, chrF, TER, COMET for each system

Score tables in Excel

With and without means and 95% confidence intervals

A sentence-level COMET spreadsheet

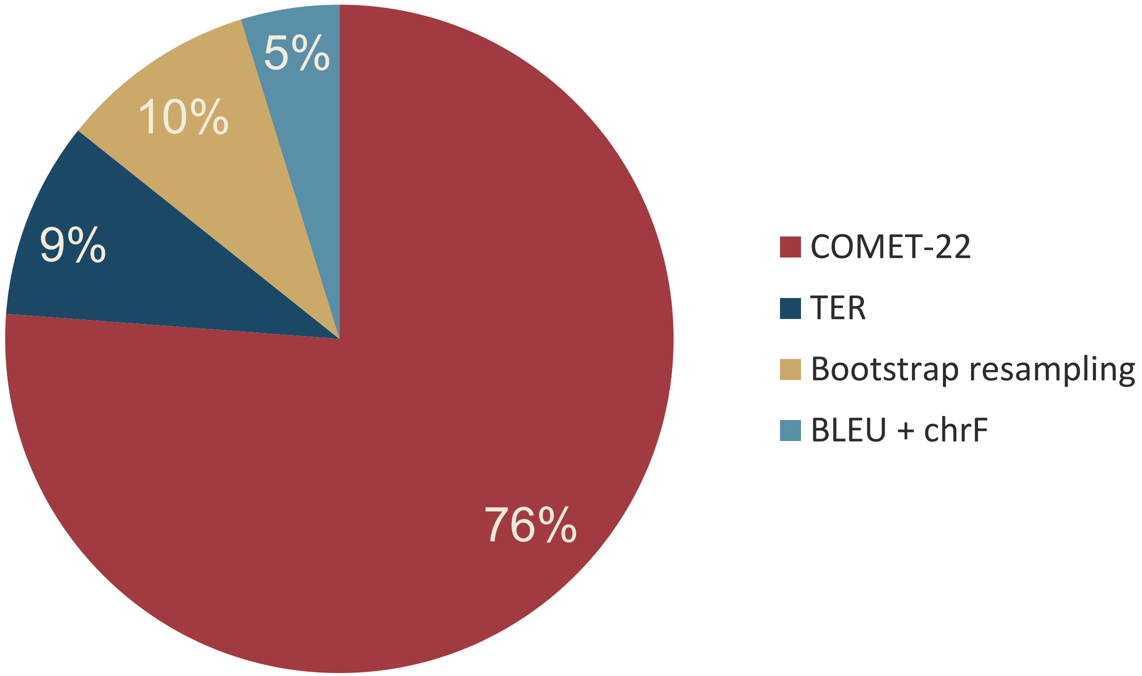
the richest output — every segment paired with every system score

Open them on your laptop, we'll go through them together.

Where the time went

On a typical run, COMET dominates total time.

~21 min total (single output, EN→DE)



WHY

- COMET runs a 600-million-parameter neural model.
- Each sentence is encoded three times (source / hypothesis / reference) and the embeddings are scored.
- TER is dynamic-programming over edits; CPU-friendly but not free.
- Bootstrap resampling computes confidence intervals — 1000+ resamples per metric (next slide).

What is “bootstrap resampling”?

How can we compute a confidence interval if we only have ONE dataset?

1

Pretend our data is the population

Treat your 300 sentences as if they perfectly represent the “universe” of similar translations.

2

Sample with replacement — many times

Draw 300 sentences randomly (with repeats allowed) from your data. Recompute the score. Repeat 1000+ times.

3

Read off the spread

Sort the 1000 scores. The 2.5th and 97.5th percentiles bound the 95% confidence interval.

…and that's why MATEO needs ≈2 minutes of “bootstrap” time at the end of every run.

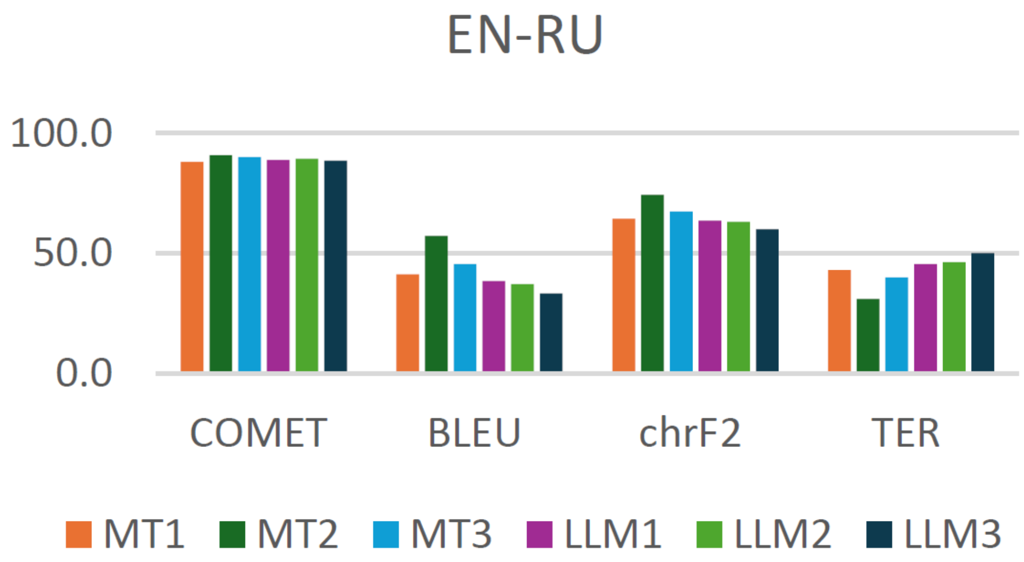
Confidence intervals — what they really mean

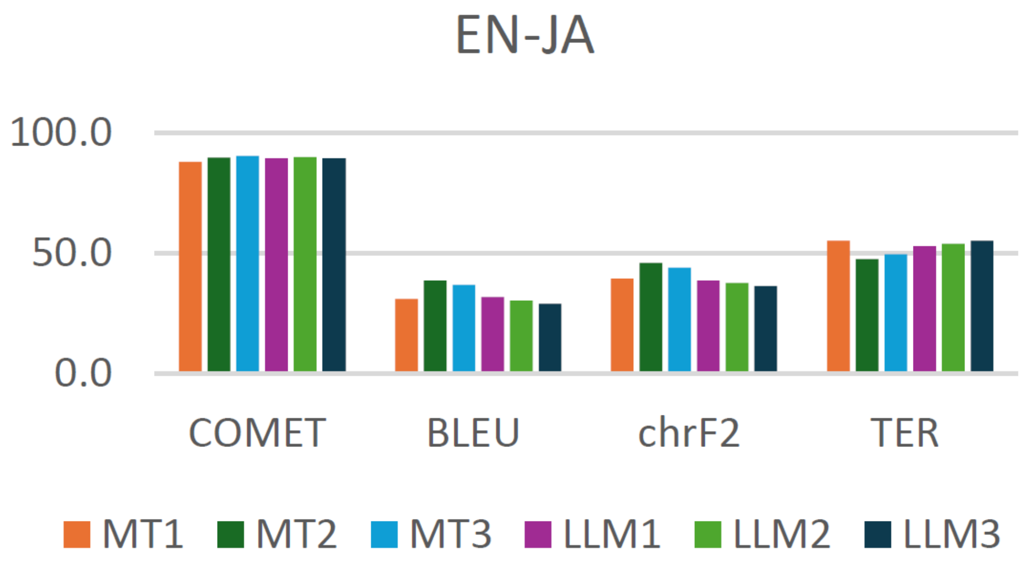
"95% CI: 86.2 ± 1.7" means...

If we ran this same evaluation on many similar 300-sentence samples, 95% of the time the true score would fall in the range 84.5 – 87.9.

Wide CI (e.g., ±5)	Score is uncertain. Could be a fluke of this particular sample. Be skeptical.
Narrow CI (e.g., ±0.5)	Score is stable. More similar runs would give similar numbers. Trust it.
Overlapping CIs across systems	Treat the ranking as inconclusive. The visible gap may not survive replication.

Reading the bar charts

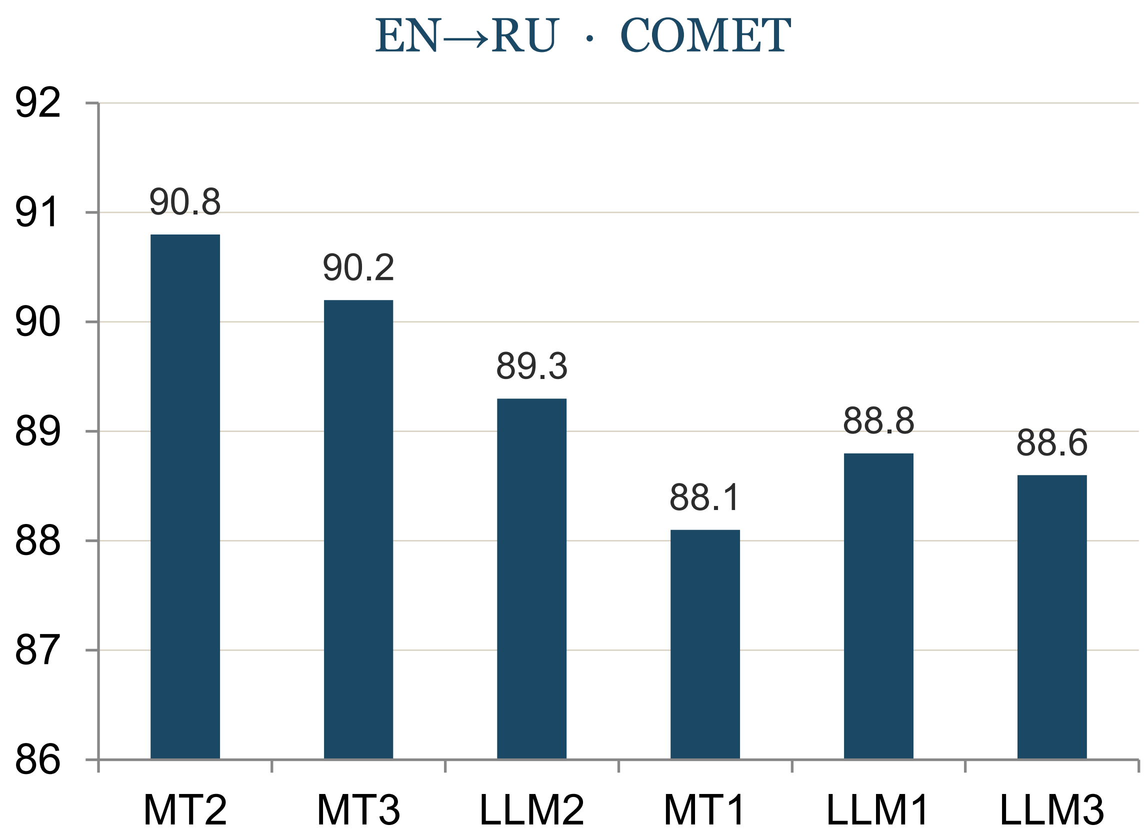


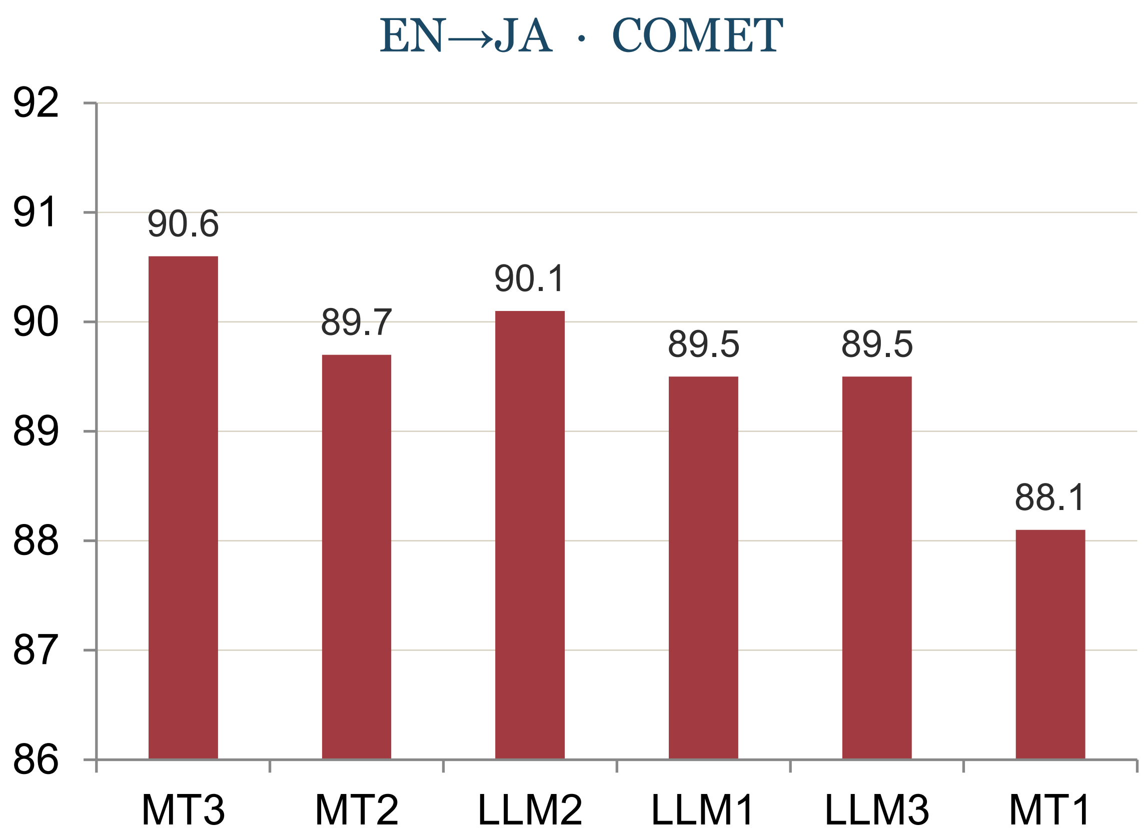


Notice: the system rankings are not identical across language pairs. MT2 leads on RU; MT3 leads on JA. The gaps are tight: most differences within ±1 COMET point.

Balashov et al., 2025

Reading the bar charts





Balashov et al., 2025

Are the differences statistically significant?

Sample size matters.

- With ≈300 segments, score differences of more than 1.5 COMET points or 3 BLEU points are usually significant.
- Smaller differences are on the cusp: may be real, may be noise.
- MATEO’s *-marked entries flag $p < 0.05$ differences from the baseline (the first system in the table).
- Want significance for tiny gaps? You need more data, or a more powerful test (Part 4).

“On the cusp” is exactly where Part 4 lives. Time to put on our statistician hats.

Sentence-level COMET in Excel

Once you have COMET per sentence, you can do a lot more than read averages.

Find the lowest-scoring outputs

Sort by COMET ascending. The bottom 5% are the disasters. Investigate them; they often reveal systematic failures.

Compare systems sentence by sentence

Where does System A win? Where does System B win? Pivot tables make this trivial.

Spot high-variance sentences

Sentences where systems disagree dramatically often expose ambiguity, OOV terms, or domain mismatch.

Pair with human grades

Pick a small set of segments to grade by hand. Correlate human and COMET; see Part 4.

Output: compile your own tables and charts

Microsoft Excel Worksheet

END OF PART 3a

Quick Q&A

Questions on Part 3?

Take 5 minutes to ask anything about MATEO outputs.

END OF PART 3b

From scores to insights

We've read the charts. We've inspected the table. We've peeked at the Excel. Now…

PART 4

Lightweight Statistical Analysis

- Mean, variance, standard deviation
- Confidence intervals and *p*-values, in plain language
- Correlation analysis: Pearson, Spearman, Kendall's τ
- Excel + a touch of LLM-assisted Python

4

Lightweight Statistical Analysis

45 minutes · the math you actually need

Why statistics for translators?

- To know when score differences mean something, and when they're noise.
- To pair automatic scores with your own human judgments rigorously.
- To talk to clients who increasingly want defensible quality numbers, not just "trust me."
- And (surprisingly often…) to generate a more confident, evidence-backed deliverable.

Common worry

*"I haven't done statistics since college." That's fine. We need only **five** (5) concepts: mean, variance, standard deviation, confidence interval, and p-value. All available as one-click formulas in Excel.*

Mean · variance · standard deviation

Mean (μ)

The average. Sum / count.

USE WHEN

Reporting an overall "how well does this system do?" score.

EXCEL

```
=AVERAGE(B2:B301)
```

Variance (σ^2)

Average squared distance from the mean.

USE WHEN

Comparing how SPREAD-OUT scores are across sentences.

EXCEL

```
=VAR.S(B2:B301)
=VAR.P(B2:B301)
```

Std. deviation (σ)

Square root of variance — same units as your data.

USE WHEN

Reporting score variation in everyday language: "COMET = 86 ± 4".

EXCEL

```
=STDEV.S(B2:B301)
=STDEV.P(B2:B301)
```

Confidence intervals — deeper

MATEO computes CIs via bootstrap. You can compute them in Excel from any sentence-level data.

- For a simple normal-approximation 95% CI: $\mu \pm 1.96 \times (\sigma / \sqrt{n})$.
- Excel: =CONFIDENCE.NORM(0.05, STDEV.S(B:B), COUNT(B:B))
- For non-normal data (and translation scores are rarely normal!), use a *bootstrap* script: easy to ask an LLM to write.
- Always ask: does my CI overlap the next system's CI? If yes, the ranking is inconclusive.

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$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\begin{aligned}
\mu &= \text{Mean} \\
\sigma &= \text{Standard Deviation} \\
\pi &\approx 3.14159\cdots \\
e &\approx 2.71828\cdots
\end{aligned}$$

Confidence intervals — deeper

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- Always ask: does my CI overlap the next system's CI? If yes, the ranking is inconclusive.

Try this prompt

"Write a 10-line Python script that reads a CSV with a 'COMET' column, draws 1000 bootstrap samples (with replacement), and prints the 2.5th and 97.5th percentiles."
- Most modern LLMs do this in seconds.

p-values — what they really mean

A p-value is the probability of seeing your data, or something more extreme, IF the null hypothesis were true.

- Null hypothesis: "there's no real difference between System A and System B."
- Small p (< 0.05): your data would be very unlikely under the null. Conclude: there probably IS a difference.
- p does NOT measure the size of the difference — only how confident we are that some difference exists.
- A statistically significant tiny difference may be practically irrelevant. Always report effect size alongside p.

Excel formulas

Paired t-test: =T.TEST(A:A, B:B, 2, 1)
Two-sample: =T.TEST(A:A, B:B, 2, 2)

(Returns the p-value directly.)

Bootstrap-style p-values for translation metrics: ask an LLM to script it.

p-values — what they really mean

A p-value is a way to ask: "Would this result be surprising if there were really no difference?"

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Think of p as a "surprise score"

Small p-value = more surprising under "no difference"

p-values — what they really mean

A p-value is a way to ask: "Would this result be surprising if there were really no difference?"

- Start by pretending the boring answer is true: **"There is no real difference"** between the quality of two MT outputs
- Then ask: how often would we see a result this strong, or stronger, just by chance?
- That chance is the *p*-value.
- *p* = 0.05 means "about 5 times out of 100" under the no-difference assumption.

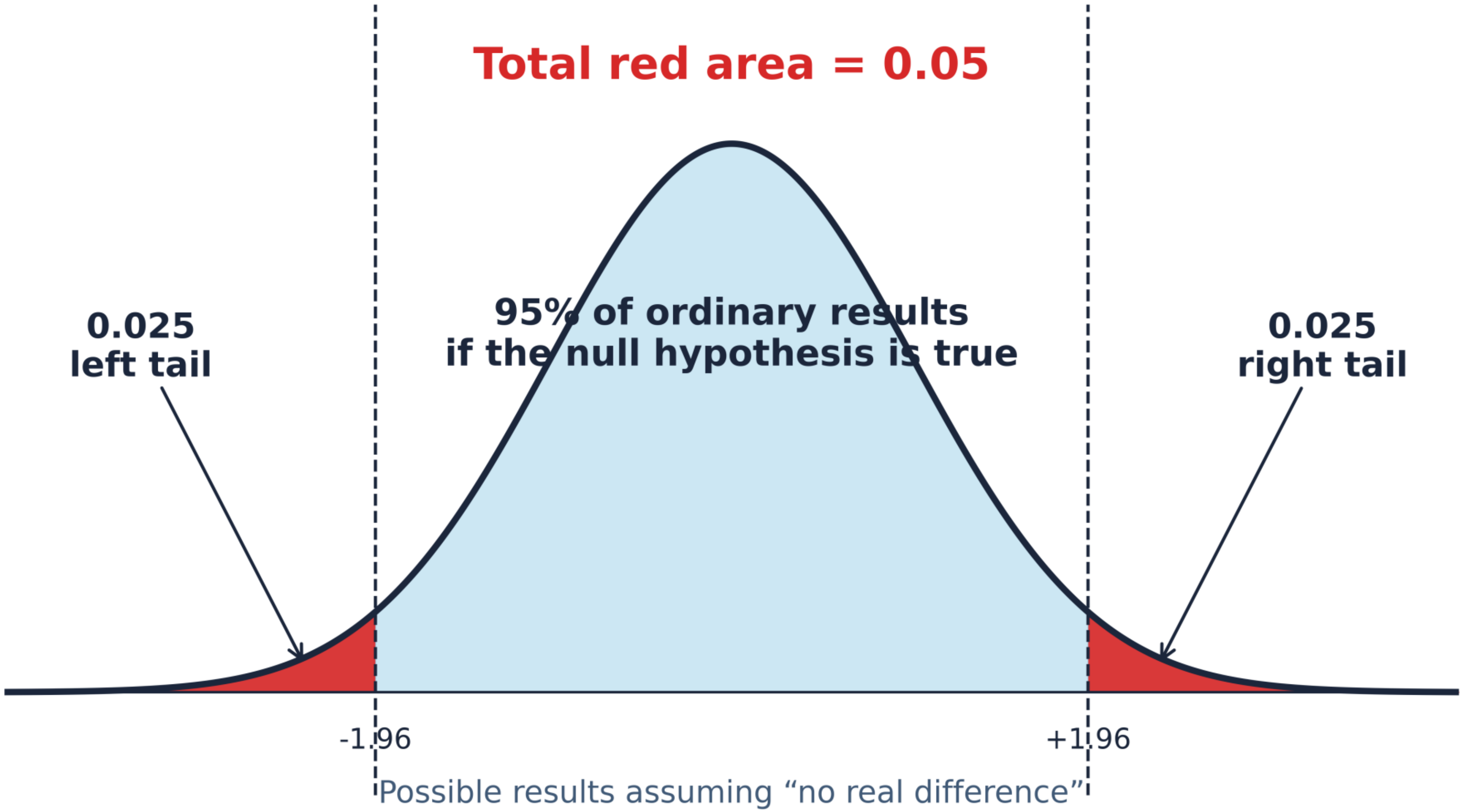
Think of *p* as a "surprise score"

Small *p*-value = more surprising under "no difference"

What p=0.05 looks like

In a two-sided test, the usual 0.05 cutoff is split into two rare tails: 0.025 + 0.025

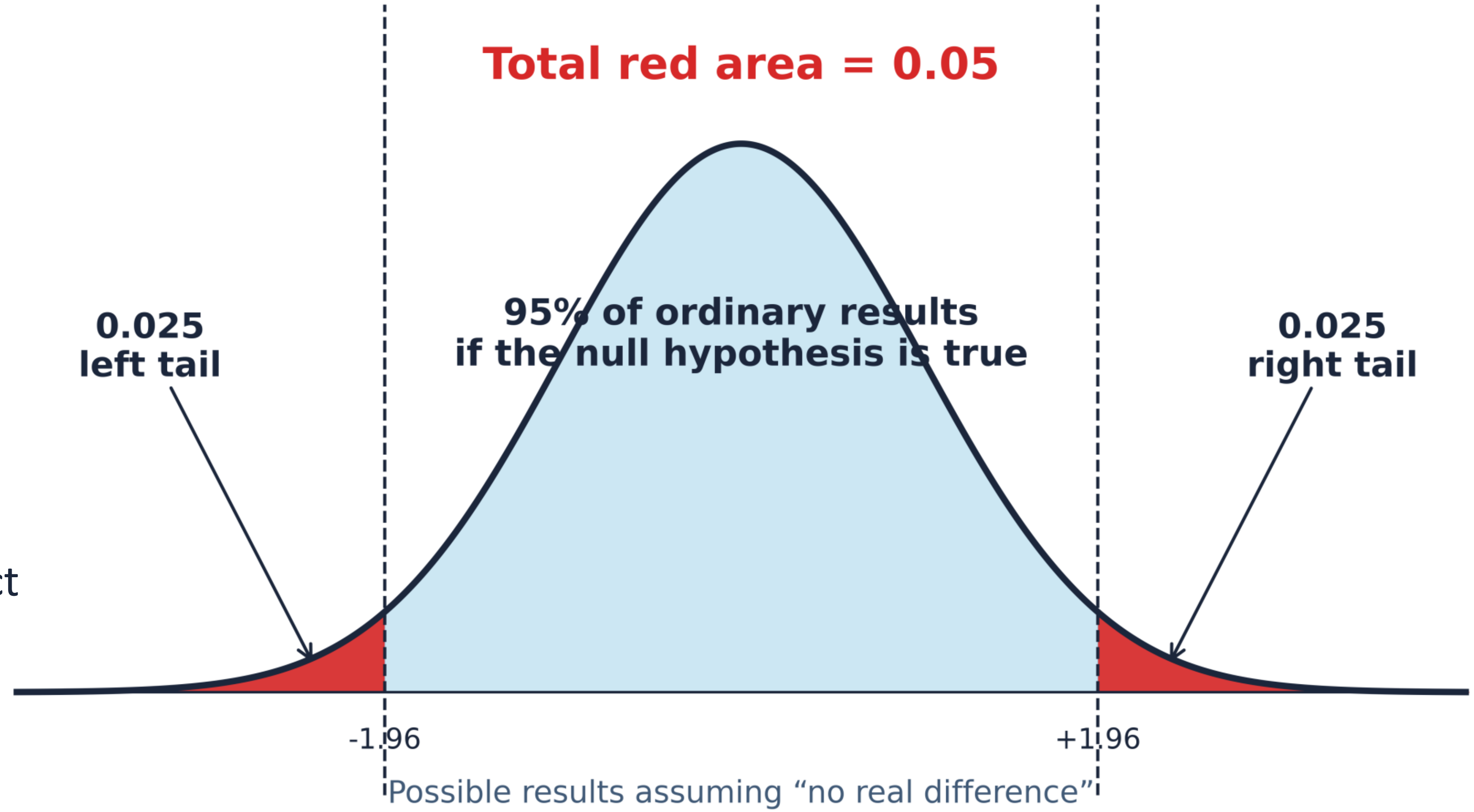
Plain English: if there really were no difference, only 5% of results would be this extreme or more extreme.



The safe way to say it

“This result would be unlikely if there were really no difference.”

Use *p*-values to judge evidence that some difference exists. Use effect size to judge whether the difference matters.



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A p-value is the probability of seeing your data, or something more extreme, IF the null hypothesis were true.

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(Returns the p-value directly.)

Bootstrap-style p-values for translation metrics: ask an LLM to script it.

Your Excel statistical toolkit

WHAT YOU WANT	EXCEL FORMULA
Average score	`=AVERAGE(range)`
Standard deviation	`=STDEV.S(range)` `=STDEV.P(range)`
95% confidence interval (½-width)	`=CONFIDENCE.NORM(0.05, STDEV.S/P(range), COUNT(range))`
Pearson correlation	`=CORREL(range1, range2)`
Spearman correlation	`=CORREL(RANK.AVG(range1,range1,1), RANK.AVG(range2,range2,1))`
Two-sample *t*-test *p*-value	`=T.TEST(range1, range2, 2, 2)`
Paired *t*-test *p*-value	`=T.TEST(range1, range2, 2, 1)`

Python — and why I keep mentioning it

Anything Excel does, Python can do, usually faster and reproducibly.

- Most popular general-purpose programming language; preferred by LLMs themselves (Twist et al., 2025).
- scipy.stats does every test discussed today, and many we won't.
- pandas reads/writes Excel transparently; you don't have to learn a new file format.
- For loops over many systems, languages, or sample sizes: Python is hours faster than clicking in Excel.
- Modern LLMs write working scripts in seconds. You don't need to be a programmer to USE Python.

Try this LLM prompt

"Read sentence_scores.xlsx in pandas. For each system, compute the mean, standard deviation, 95% bootstrap CI (1000 samples), and the Pearson correlation with the human_grade column. Print as a markdown table."

GPT-5 / Claude / Gemini all do this in one shot.

Correlation analysis: introduction

Two questions correlation answers:

Do automatic metrics agree?

If chrF says System A wins, does COMET agree? Strong inter-metric correlation = consensus.

Do automatic scores reflect human judgment?

If a sentence got COMET 0.92, does a human evaluator also rate it highly? This is the holy grail.

Pearson · Spearman · Kendall's τ

Pearson r *Linear association.*	USE IT WHEN When relationships are roughly linear and data are continuous (COMET vs. BLEU).
Spearman ρ *Rank-based association.*	USE IT WHEN When you care about order, not magnitude (high-COMET ↔ high-human-grade?).
Kendall's τ *Concordant vs. discordant pairs.*	USE IT WHEN Robust to outliers and ties. Slower to compute, but the gold-standard rank measure.

All three return values in [-1, +1]. Sign tells direction; magnitude tells strength. ±0.7 is strong; ±0.3 weak.

Pearson · Spearman · Kendall's τ

Pearson r

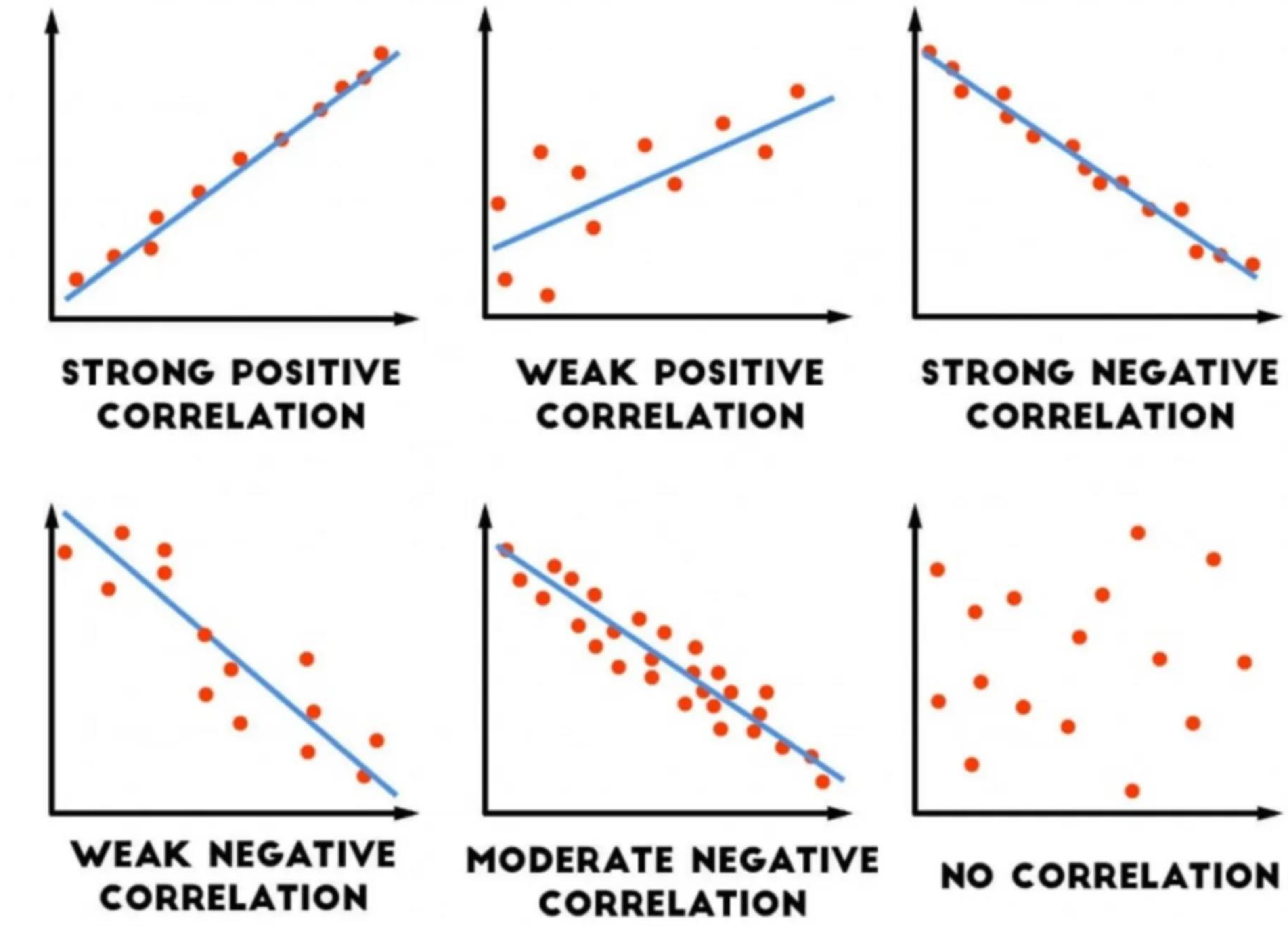
Linear association.

USE IT WHEN

When relationships are roughly linear and data are continuous (COMET vs. BLEU).

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

How much do two variables change together relative to how much they change Individually?



https://medium.com/upskilling/data-visualization-part-7-correlations-differentiation-and-linear-regression-d1474bd2f3f2

Document-level metric × metric

Pairwise Pearson correlations across our six MT/LLM systems (Reeve corpus).

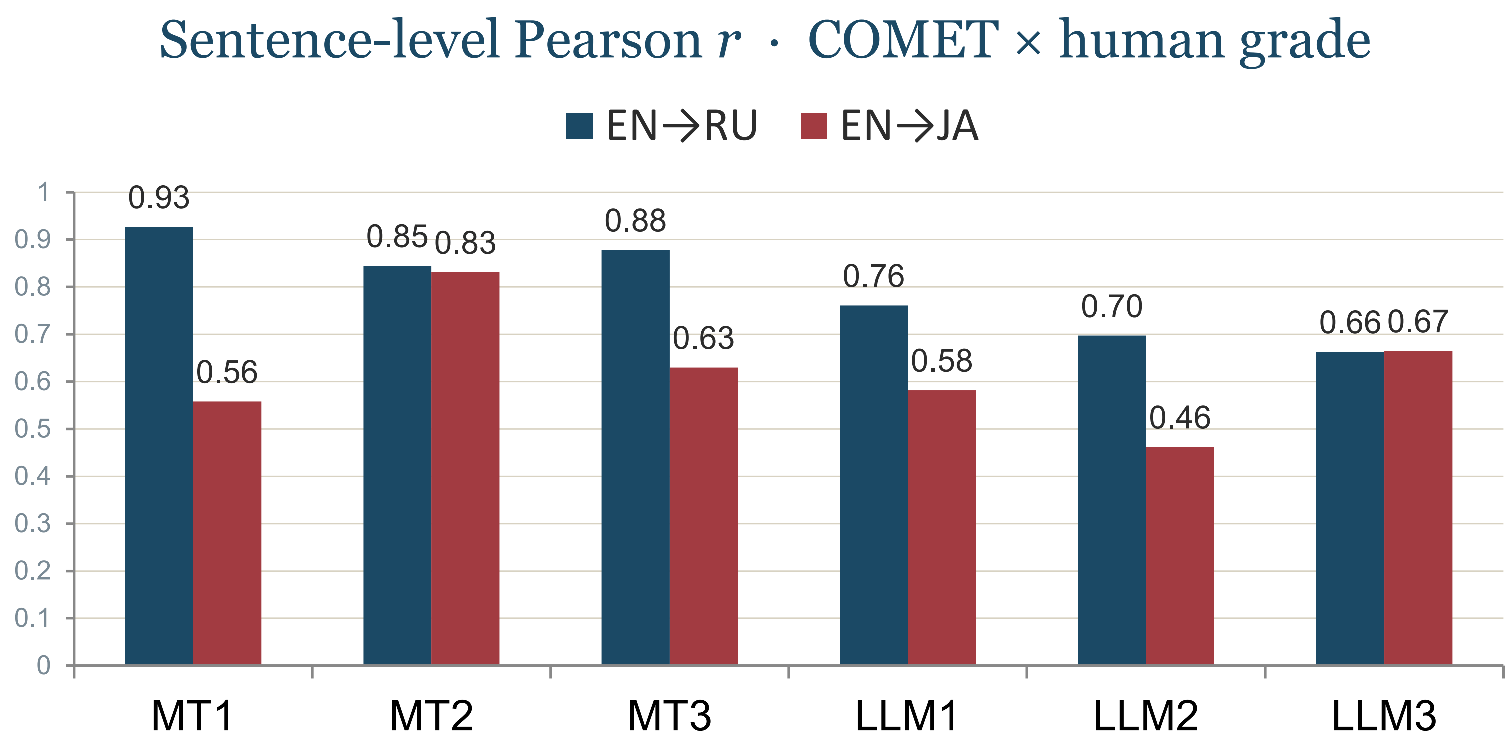
PAIR	EN→RU	EN→JA	INTERPRETATION
BLEU vs chrF	0.998	0.990	Very strong: these two surface metrics largely agree.
BLEU vs TER	−0.999	−0.969	Almost perfect inverse: TER is essentially “BLEU upside down.”
BLEU vs COMET	0.806	0.388	Moderate for RU; low for JA: COMET sees what BLEU doesn't.

THE KEY MOMENT

BLEU and COMET correlate strongly for EN→RU but barely correlate for EN→JA. The Japanese case is where neural metrics earn their keep: they capture quality dimensions that string overlap cannot.

Sentence-level: COMET vs. human grades

Pearson correlations between sentence-level COMET and our human grades (0.0–4.0).



READING THE BARS

- **EN→RU: r ≈ 0.66–0.93**
  moderate-to-strong agreement
- **EN→JA: r ≈ 0.46–0.83**
  noticeably weaker: COMET still helpful but sentence-level scores are noisier on JA
- **Practical lesson**
  trust COMET system-level rankings; treat individual sentence scores as suggestive, not definitive.

Human evaluation is a world of its own

What we're doing today is the simplest end of a deep field.

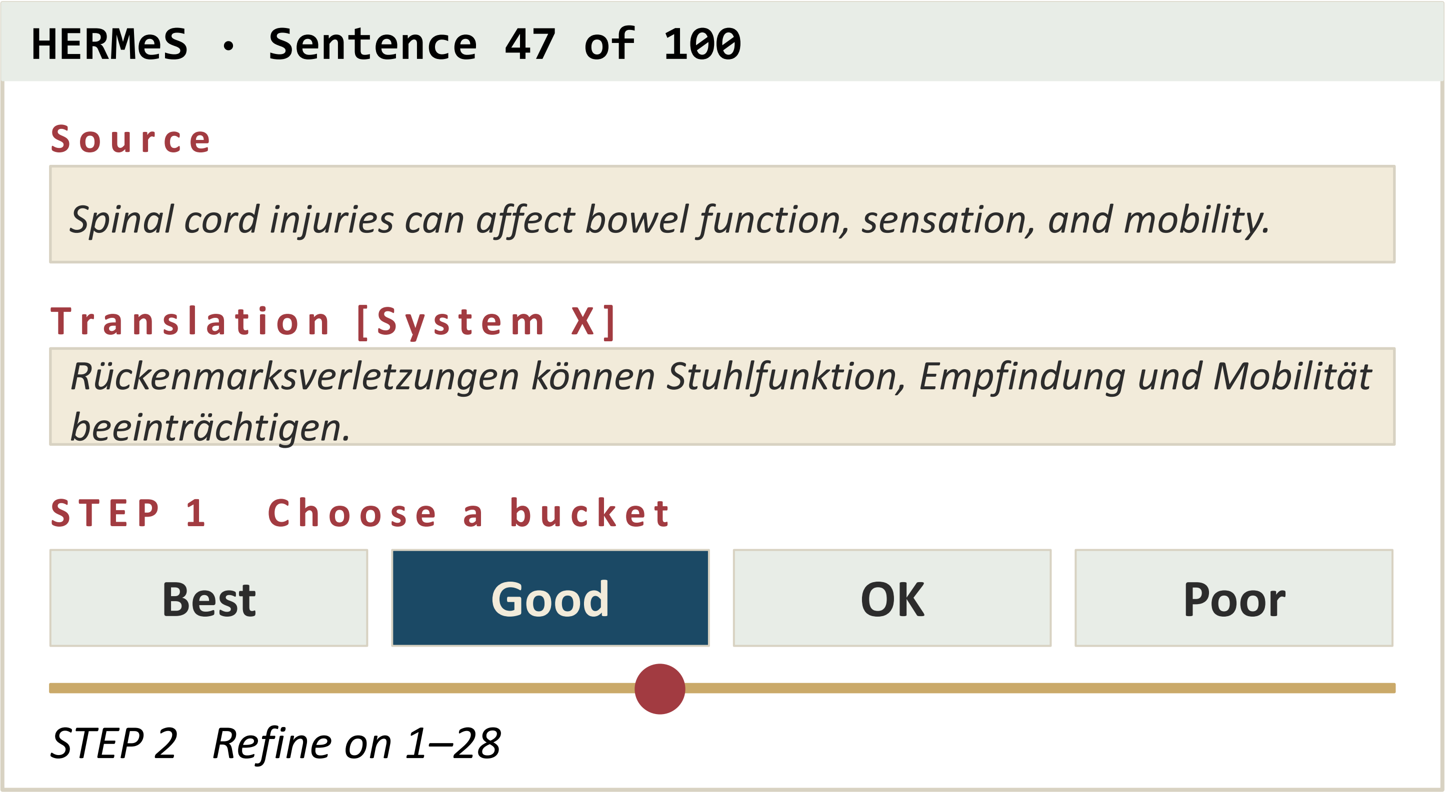
- Inter-annotator agreement (κ, α, ρ): when do two evaluators agree?
- Annotator calibration: drift over time, fatigue effects.
- Score normalization: z-scores per annotator to remove individual harshness.
- Variance-based and quality-estimation-informed sampling (Zouhar et al., 2025).
- MQM campaigns: Freitag et al. (2021, 2022, 2024) at WMT.

Going further

Mukherjee & Shrivastava (2026), "Lost in Translation? Found in Evaluation" — a recent comprehensive survey on sentence-level translation evaluation. Linked from the tutorial bibliography.

Lightweight human evaluation

A Streamlit interface developed by Rex VanHorn (UGA) for our recent variance-based studies.



What it gives you

- Single integer score per segment
- Built-in randomization & blind presentation
- Cryptographic submission hashes
- Output: tidy CSV ready for correlation analysis

Open source: adaptable to your own evaluations.

https://mt-eval.streamlit.app/?type=admin

https://github.com/RexVH/St-MT-Evaluation-app

Analyzing human × COMET correlations

You now have everything you need to evaluate any new MT system on your own data.

- Pick 30–60 sentences spanning the COMET range (high, medium, low).
- Grade them with HERMeS (or any consistent rubric).
- In Excel: =CORREL(human_grades, comet_scores) — Pearson *r* in one cell.
- Interpret: $r > 0.6$ = COMET tracks your judgment well; $r < 0.3$ = COMET and you disagree on this domain.
- Repeat for new domains, new clients, new engines. This is how you build defensible quality assessments.

END OF PART 4

Part 4 recap

Five concepts. Three formulas. One workflow.

- Mean and standard deviation describe a system's overall quality.
- Confidence intervals tell you whether differences are real or noise.
- *p*-values tell you the strength of evidence (not the size of the effect!).
- Pearson, Spearman, and Kendall measure how well two scores travel together.
- All of these are one Excel formula away; or a 10-second LLM prompt.

From here

Try it on your next project. Grade ten segments, run the correlations, see what you find. The first time will take an hour; by the third project, it will be twenty minutes; plus: you'll have a quantitative quality story to tell your client.

Key takeaways

1. **Translation evaluation is no longer just for big tech.**
MATEO + a browser puts COMET-grade rigor in your daily workflow.

2. **COMET is the default neural metric.**
BLEU, chrF, and TER are valuable cross-checks, not the headline numbers.

3. **Confidence intervals tell you what the means hide.**
If two systems' CIs overlap, the visible gap may be statistical noise.

4. **Sentence-level data is where the gold is.**
MATEO's Excel export is your most powerful artifact; go beyond document averages.

5. **Translation analytics is a billable, differentiating skill.**
Clients increasingly want defensible quality numbers. You can now provide them.

New skills, new opportunities

Where to go from here

Pilot it on your next project

Pick a job with a TM and 2–3 candidate engines. Run an evaluation. Decide with data.

Offer it as a service

Translation analytics as a discrete deliverable: short report, charts, recommendation. Bill accordingly.

Keep learning

Read up. Track the ATA Chronicle and Slator. Watch the EAMT/AMTA conference programs.

Join the community

EAMT, AMTA, ATA, and our growing community of translator-analysts. We're building this together.

Questions?

Final Q&A: 10 to 15 minutes

Anything from any of the four parts. The floor is yours.

Thank you.

All materials — slides, evaluation sets, scripts, bibliography — are openly available.

GitHub repository

github.com/YuriBalashov/eamt2026-eval-tutorial

Slides · evaluation sets · README · bibliography · worked examples · Excel templates

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